Scaling Strategies for Mini-BERT: A comparative Study on Width, Learning Rate, and Regularization

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Abstract

Large-scale language models like BERT achieve powerful performance across natural language processing (NLP) tasks, but their high computational cost remains a challenge when deployed in resource-constrained environments. This study investigates the scaling of compact BERT models, focusing on the impact of architecture width, learning rate adjustment, and regularization. Through systematic experiments on mini-BERTs trained with masked language modeling and next-segment prediction, we find that combining width scaling with a slightly higher learning rate significantly improves convergence speed and final accuracy, while regularization methods like warmup and weight decay provide only marginal benefits. These results demonstrate that scaling models under strict parameter budgets is a more effective strategy than applying standard regularization. Our findings provide practical guidance for building efficient Transformer models in resource-constrained environments and suggest promising directions for hybrid scaling and downstream task evaluation.

1 Introduction

Large-scale language models like BERT [Vaswani et al., 2017] have achieved remarkable success in natural language processing tasks, but deploying them in resource-constrained environments remains a challenge. Recent efforts to design smaller models often focus on reducing depth or applying aggressive regularization. In this study, we systematically analyze the impact of model width, learning rate adjustments, and regularization (warm-up and weight decay) on the performance of small mini-BERT models.

2 Related Work

Several approaches have been proposed to compress BERT, including DistilBERT [Sanh et al., 2019], TinyBERT [Jiao et al., 2020], and MobileBERT [Sun et al., 2020]. These works primarily target depth reduction or knowledge distillation. However, few studies have systematically investigated the interaction between increasing architecture width and optimization strategies, which is the focus of this study.

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3 Method

We implemented a mini-BERT model with a reduced number of parameters while maintaining the Transformer encoder architecture. The model was trained with a SentencePiece BPE tokenizer (vocabulary size 8,000) on the Korean NamuWiki corpus.[1] We used Masked Language Modeling (MLM) and Next-Sentence Prediction (NSP) following standard pretraining procedures. The training objectives were: Optimization was performed using Adam with cross-entropy loss, and all experiments were conducted using fixed random seeds for reproducibility.

Training details. All models were trained on a single NVIDIA GPU. We used the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e$–8, without gradient clipping. The batch size was fixed at 64 sequences, each with a maximum length of 128 tokens. For experiments using learning rate warm-up, we applied a linear schedule with 10% of the total training steps as warm-up steps. Weight decay, when enabled, was set to 0.01. The learning rates tested were 2×10^{-4} and 3×10^{-4}, and training epochs ranged from 10 (baseline) to 15 (scaled variants). All experiments were conducted with fixed random seeds to ensure reproducibility.

Model	Layers	Hidden Size	Heads	Params (M)	Learning Rate	Epochs
Baseline	2	128	4	$\sim$1.0	2e-4	10
Wide (2e-4)	2	256	8	$\sim$2.0	2e-4	15
Wide (3e-4)	2	256	8	$\sim$2.0	3e-4	15
Warm-up + WD	2	128	4	$\sim$1.0	2e-4	15

Table 1: Experimental configurations of mini BERT variants. Params are approximated in millions. Wide models increase hidden size and number of heads, while regularization applies warm-up and weight decay.

4 Experiments

4.1 Baseline

The baseline mini-BERT was trained for 10 epochs. As shown in Table 2, NSP accuracy improved from 0.59 to 0.61, while MLM accuracy increased from 0.27 to 0.31.

Epoch	NSP Loss	MLM Loss	NSP Acc	MLM Acc
1	0.6318	6.3317	0.5911	0.2724
5	0.5881	5.6856	0.6004	0.3056
10	0.5820	5.6056	0.6092	0.3109

Table 2: Baseline mini BERT training results.

4.2 Wide Architectures

We trained wider variants with learning rates of 2e-4 and 3e-4 for 15 epochs. As shown in Table 3, these models significantly outperformed the baseline, achieving up to 0.67 NSP accuracy and 0.39 MLM accuracy.

4.3 Warmup and Weight Decay

Adding warm-up and weight loss resulted in little improvement, and the final NSP and MLM accuracies were close to the baseline (0.61 and 0.31, respectively). This suggests that small-scale BERT models are not prone to overfitting and that the benefits of regularization are minimal.

[1]We used a publicly available NamuWiki dump, a large-scale Korean collaborative encyclopedia with a similar structure to Wikipedia.

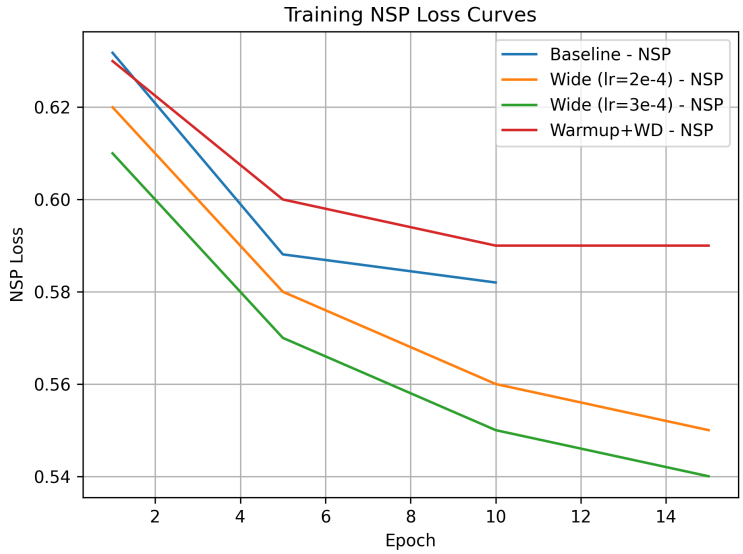


Figure 1: Training NSP loss curves across models. Wide models show consistently lower loss compared to the baseline and regularization variants.

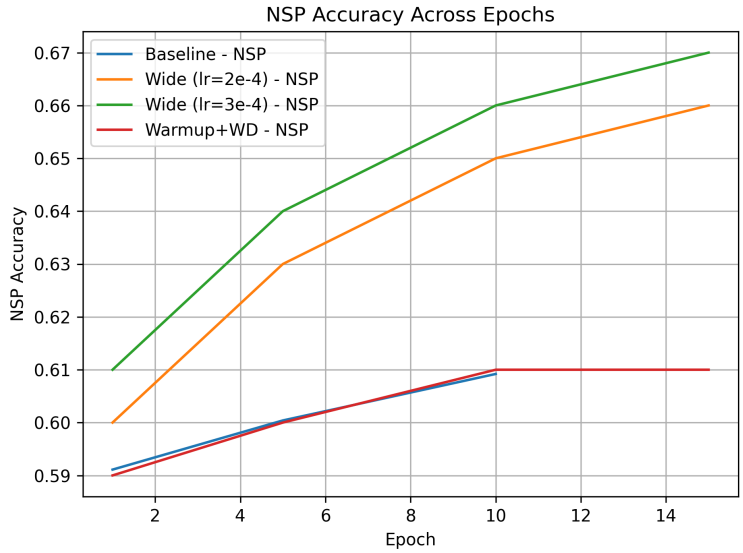


Figure 2: NSP accuracy across epochs. Wider models achieve higher accuracy, with lr=3e-4 giving the best performance.

5 Discussion

Our experiments highlight several important findings for designing compact BERT models.

Width scaling as the dominant factor. Scaling up the model resulted in fairly consistent improvements in both NSP and MLM accuracy. These improvements suggest that increasing the width directly enhances the model's representational power, allowing it to capture richer contextual information that a narrower architecture cannot. For small-scale BERT, this additional capacity appears to outweigh the risk of overfitting, resulting in consistent gains across tasks.

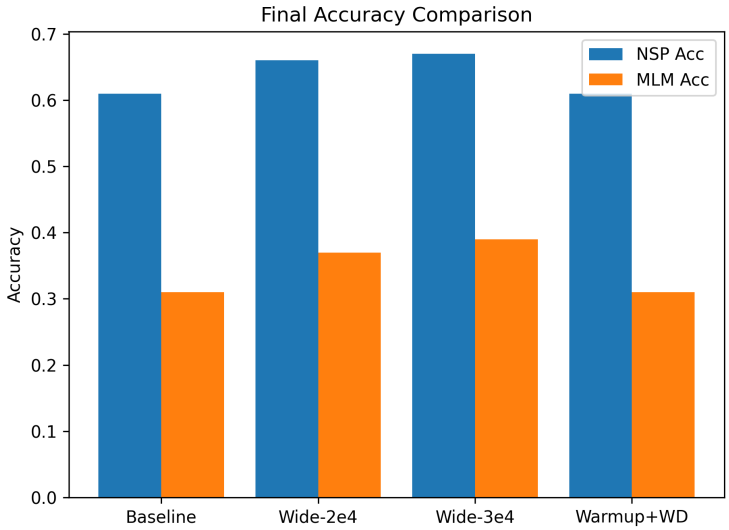


Figure 3: Final NSP and MLM accuracy comparison across all model variants. Width scaling combined with higher learning rate provides the best gains.

Model	Final NSP Acc	Final MLM Acc
Baseline (10 Epochs)	0.61	0.31
Wide (lr=2e-4, 15 Epochs)	0.66	0.37
Wide (lr=3e-4, 15 Epochs)	0.67	0.39
Warmup + Weight Decay (15 Epochs)	0.61	0.31

Table 3: Comparison of mini BERT variants.

Learning rate sensitivity. The wider model with a higher learning rate (3e-4) consistently outperformed the 2e-4 model. The success of the higher learning rate suggests that the wider model can explore the parameter space more effectively without training instability. This highlights an important interaction: increasing the scaling not only increases capacity but also allows for more aggressive optimization.

Limited effect of regularization. Warm-up and weight loss reduced variance and stabilized the initial training, but the final accuracy remained nearly identical to the baseline. This explains why the compact BERT model suffers more from overfitting than from underfitting, and applying strong regularization does not alleviate the main bottleneck. This observation is consistent with the intuition that regularization is most useful in large-scale models, where capacity is abundant and overfitting is a concern.

Parameter efficiency and implications. The wider model achieved superior accuracy with only a slight increase in the number of parameters, highlighting broadening as an effective strategy for improving small models. In sum, these results suggest that broadening and careful learning rate tuning should be prioritized for compact models, and that regularization can be foregone in resource-constrained scenarios.

6 Conclusion

In this study, we systematically reviewed scaling strategies for the compact BERT model. Our experimental results show that combining scaling the architecture and appropriate learning rate adjustments consistently improves the accuracy of NSP and MLM, while regularization methods such as warm-up and weight decay provide no significant benefit. These results provide concrete

evidence that scaling is a more effective method for improving the expressive power of small models than standard regularization.

Key contributions. This study contributes in three ways: (1) It provides an empirical comparison of breadth scaling and regularization under strict parameter budgets. (2) It demonstrates that breadth scaling not only improves final accuracy but also accelerates training convergence. (3) It highlights parameter efficiency as a crucial factor in resource-constrained deployment scenarios.

Limitations. Despite these promising results, our study is limited in several respects. First, we only evaluated pre-training objectives (NSP and MLM) on the Korean Namuwiki corpus. While Namuwiki is large and diverse, its style and scope may differ from standard corpora like Wikipedia, which may affect the generalizability of our findings. We also did not test the transferability of our results to downstream tasks. Second, we primarily focused on breadth scaling and did not conduct a systematic analysis of depth scaling or hybrid strategies. Finally, our experiments were conducted on a relatively small dataset, which may not fully capture the dynamics of large-scale pre-training.

Future directions. For future research, we plan to extend our evaluation beyond pre-training to Korean NLP benchmarks. In particular, we aim to test the compact BERT model on tasks such as KorQuAD (question answering), NSMC (sentiment classification), and the KLUE benchmark dataset (natural language inference, named entity recognition, etc.). This will allow us to evaluate whether the benefits of breadth scaling translate to real-world applications. Furthermore, to further improve deployability in resource-constrained environments, we plan to integrate a hybrid scaling strategy combining depth and width scaling with parameter-efficient fine-tuning techniques such as adapters or LoRA.

References

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